Problem 6

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Problem 1 *Find the critical points of f on the given interval. Determine whether the function f has a local maximum, local minimum or no local extremum at each critical point.* ***EXPLAIN****.*

(a) $f(x) = x\sqrt{2-x^2}$ *on* $(-\sqrt{2}, \sqrt{2})$.

Explanation.

$$\begin{aligned} f'(x) &= \sqrt{2-x^2} + x\left(\frac{1}{2\sqrt{2-x^2}}(-2x)\right) \\ &= \sqrt{2-x^2} - \frac{x^2}{\sqrt{2-x^2}} \\ &= \frac{2-x^2-x^2}{\sqrt{2-x^2}} \\ &= \frac{2(1-x^2)}{\sqrt{2-x^2}} \end{aligned}$$

Critical points of f occur where $f'(x) = 0$ or where $f'(x)$ does not exist. Solving $f'(x) = 0$ yields that $2(1-x^2) = 0$, or $x = \pm 1$. $f'(x)$ does not exist when $2 - x^2 \leq 0$, but the points $x = \pm\sqrt{2}$ are not in the domain.

Let us use the Second Derivative Test to determine whether the function f has a local maximum or local minimum at the points where $x = \pm 1$.

$$f''(x) = \frac{d}{dx}\frac{2(1-x^2)}{\sqrt{2-x^2}} = \frac{2x(x^2-3)}{(2-x^2)^{\frac{3}{2}}}$$

Since $f''(-1) = 4$, the point $(-1,-1)$ is a local minimum by the Second Derivative Test. Since $f''(1) = -4$, the point $(1,1)$ is a local maximum by the Second Derivative Test.

(b) $f(x) = x^3e^{-x}$ *on* $(-1, 5)$.

Explanation.

$$\begin{aligned} f'(x) &= 3x^2e^{-x} + x^3(-e^{-x}) \\ &= x^2e^{-x}(3-x) \end{aligned}$$

Notice that $f'(x)$ always exists, and so all of the critical points of f occur when $f'(x) = 0$. Solving this equation:

$$x^2e^{-x}(3-x) = 0$$

$$x^2(3-x) = 0$$

$$x = 0 \qquad \text{or} \qquad x = 3$$

It is easier to use the First derivative test in order to determine whether the function f has a local maximum, local minimum or no local extremum at the point where $x = 0$. Since $-1 < 0 < 1$, $f'(-1) = 4e > 0$, and $f'(1) = 2e^{-1} > 0$, it follows that the sign of f' does not change at $x = 0$. The function f has **no local extremum** at $x = 0$.

Similarly, we will use the First derivative test in order to determine whether the function f has a local maximum, local minimum or no local extremum at the points where $x = 3$. Since $1 < 3 < 4$, $f'(1) = 2e^{-1} > 0$, and $f'(4) = -16e^{-4} < 0$ it follows that the sign of f' changes from positive to negative at $x = 3$. The function f has **a local maximum** at $x = 3$ by the First Derivative Test.

(c) $f(x) = x\ln\left(\frac{x}{5}\right)$ *on* $(0, 5)$.

Explanation.

$$\begin{aligned} f'(x) &= \ln\left(\frac{x}{5}\right) + x \cdot \frac{5}{x} \cdot \frac{1}{5} \\ &= \ln\left(\frac{x}{5}\right) + 1 \end{aligned}$$

Notice that $f'(x)$ exists for all values in $(0, 5)$, and so all of the critical points of f occur when $f'(x) = 0$. Solving this equation:

$$\ln\left(\frac{x}{5}\right) + 1 = 0$$

$$\ln\left(\frac{x}{5}\right) = -1$$

$$\frac{x}{5} = e^{-1}$$

$$x = 5e^{-1} = \frac{5}{e}$$

Let's use the Second Derivative Test to determine whether the function f

has a local maximum, local minimum or no local extremum at the points where $x = \frac{5}{e}$.

$$\begin{aligned} f''(x) &= \frac{d}{dx}(\ln\left(\frac{x}{5}\right) + 1) \\ &= \frac{d}{dx}(\ln(x) - \ln(5) + 1) = \frac{1}{x} \end{aligned}$$

Since, $f''\left(\frac{5}{e}\right) > 0$, the point $\left(\frac{5}{e}, -\frac{5}{e}\right)$ is a local minimum by the Second Derivative Test.